

P. E. S. COLLEGE OF ENGINEERING, MANDYA

(Affiliated to VTU, Belagavi | Autonomous – UGC | NAAC 'A' Grade | NBA Accredited)

K V Shankaragowda Rd, PES College Campus, Mandya, Karnataka – 571 401

DEPARTMENT HANDBOOK

Department of Electronics & Communication Engineering

Academic Year 2025 – 2026

Established 1967 | B.E. (EC) – Intake: 180 | M.Tech (VLSI & ES) – Intake: 18 | Ph.D.

1. About the Department

The Department of Electronics and Communication Engineering (ECE) at P.E.S. College of Engineering, Mandya started the B.E. degree programme in 1967 and has since grown into one of the key departments at the institution. The M.Tech. programme in VLSI Design and Embedded Systems was introduced in 2006, and the Ph.D. and M.Sc. (by research) programmes were started in 2004. The undergraduate programme initially had an intake of 60 students, which has now expanded to 180 students per year.

The department is focused on providing high-quality technical education in the field of electronics and communication. It comprises highly qualified and experienced teachers, supporting staff, well-equipped laboratory facilities, staff rooms with internet connectivity, a seminar room, a library, a computer centre, and other infrastructural facilities. The location and atmosphere are conducive to learning.

The curriculum is designed to cover a wide array of topics in electronics and communication, such as VLSI, embedded systems, signal processing, communication systems, and IoT. The department emphasises research-driven education, aiming to make students industry-ready through practical exposure and projects. Faculty members hold Ph.D. degrees and have significant research and industry experience; they are involved in cutting-edge research and publish regularly in reputed journals.

1.1 Department at a Glance

Department Name	Electronics & Communication Engineering (ECE)
UG Programme Started	1967
PG Programme Started	2006 (M.Tech – VLSI & Embedded Systems)
Research Programmes	Ph.D. and M.Sc. (by Research) – started 2004
Parent Institution	P.E.S. College of Engineering, Mandya (Est. 1962)
Affiliation	Visvesvaraya Technological University (VTU), Belagavi
Also Recognised by	University of Mysuru (UOM) as Research Centre
Autonomous Status	UGC Autonomous (2008-09)
Accreditation	NAAC – A Grade NBA Accredited
UG Intake (B.E.)	180 students per year
PG Intake (M.Tech)	18 students per year
UG Programme Duration	4 Years
PG Programme Duration	2 Years
Department Page	https://pesce.ac.in/dep-electronics-communication-engg.php

1.2 NBA Accreditation History

Programme Name	Duration	Accreditation Start	Accreditation End
Electronics and Communication Engineering	3 Years	02-12-2017	30-06-2020
Electronics and Communication Engineering	3 Years	28-05-2020	30-06-2023
Electronics and Communication Engineering	3 Years	30-06-2023	30-06-2026

2. Vision, Mission & Programme Objectives

2.1 Vision

"The department of E & C would endeavour to create a pool of Engineers who would be extremely competent technically, ethically strong and also fulfil their obligation in terms of social responsibility."

2.2 Mission

The Department of Electronics and Communication Engineering is committed to:

Code	Mission Statement
DM1	Adopt the best pedagogical methods and provide the best facility, infrastructure and an ambience conducive to imbibe technical knowledge and practise ethics.
DM2	Group and individual exercises to inculcate the habit of analytical and strategic thinking to help students develop creative thinking and instil team skills.
DM3	MoUs and sponsored projects with industry and R & D organisations for collaborative learning.
DM4	Enabling and encouraging students for continuing education and moulding them for a life-long learning process.

2.3 Programme Educational Objectives (PEOs)

Graduates of the programme will be able to:

PEO	Objective
PEO 1	Exhibit knowledge in mathematics and engineering fundamentals applied to Electronics and Communication Engineering for professional achievement in industry, research, and academia.
PEO 2	Identify, analyse, and apply engineering concepts for the design of Electronics and Communication Engineering systems and demonstrate multidisciplinary expertise to handle societal needs and meet contemporary requirements.
PEO 3	Perform with leadership qualities, team spirit, management skills, attitude, and ethics needed for a successful career, sustained learning, and entrepreneurship.

2.4 Programme Outcomes (POs)

Engineering graduates will be able to:

PO	Description
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PO1 – Engineering Knowledge	Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialisation.
PO2 – Problem Analysis	Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
PO3 – Design/Development of Solutions	Design creative solutions for complex engineering problems and design/develop systems, components, or processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society, and environment.
PO4 – Investigations	Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis, and interpretation of data to provide valid conclusions.
PO5 – Modern Tool Usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, recognising their limitations to solve complex engineering problems.
PO6 – Engineer and Society	Analyse and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture, and environment.
PO7 – Ethics	Apply ethical principles and commit to professional ethics, human values, diversity, and inclusion; adhere to national and international laws.
PO8 – Teamwork	Function effectively as an individual, and as a member or leader in diverse and multidisciplinary teams.
PO9 – Communication	Communicate effectively and inclusively within the engineering community and society, including writing effective reports and design documentation, and making effective presentations.
PO10 – Project Management & Finance	Apply knowledge and understanding of engineering management principles and economic decision-making to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.
PO11 – Life-Long Learning	Recognise the need for, and have the preparation and ability for independent and life-long learning, adaptability to new and emerging technologies, and critical thinking in the broadest context of technological change.

2.5 Programme Specific Outcomes (PSOs)

After graduation, the student will have:

PSO	Outcome
PSO 1	An ability to understand the basic concepts in Electronics & Communication Engineering and to apply them in the design and implementation of Electronics and Communication systems.
PSO 2	An ability to solve complex problems in Electronics and Communication Engineering using the latest hardware and software tools, along with analytical skills to arrive at appropriate solutions.

3. Academic Programmes & Syllabus

3.1 Programmes Offered

Programme	Degree	Intake	Duration
Electronics & Communication Engineering (EC)	B.E.	180	4 Years
VLSI Design and Embedded Systems (M.Tech – VLSI & ES)	M.Tech	18	2 Years
Doctor of Philosophy	Ph.D.	–	3–5 Years
M.Sc. by Research	PG	–	2 Years

3.2 Programme Highlights

- UG focus areas: Communication, Signal Processing, VLSI Design, Embedded Systems, Image Processing, IoT.
- M.Tech specialisation: VLSI Design and Embedded Systems (introduced 2012).
- Research centre recognised by VTU, Belagavi and University of Mysuru.
- Part-time and full-time Ph.D. programmes available.

3.3 Syllabus Documents

Document	Scheme	Link
I Year Syllabus	2021 Scheme	https://pesce.ac.in/admin/photo/6/syllabuspdf/133.pdf
II Year Syllabus	2021 Scheme	https://pesce.ac.in/admin/photo/6/syllabuspdf/134.pdf
III Year Syllabus	2021 Scheme	https://pesce.ac.in/admin/photo/6/syllabuspdf/135.pdf
IV Year Syllabus	2021 Scheme	https://pesce.ac.in/admin/photo/6/syllabuspdf/138.pdf
I Year Syllabus	2022 Scheme	https://pesce.ac.in/admin/photo/6/syllabuspdf/139.pdf
II Year Syllabus	2022 Scheme	https://pesce.ac.in/admin/photo/6/syllabuspdf/140.pdf
III Year Syllabus	2022 Scheme	https://pesce.ac.in/admin/photo/6/syllabuspdf/157.pdf
IV Year Syllabus	2022 Scheme	https://pesce.ac.in/admin/photo/6/syllabuspdf/142.pdf
II Year Syllabus	2024 Scheme	https://pesce.ac.in/admin/photo/6/syllabuspdf/160.pdf
PG I & II Year Syllabus	2022 Scheme	https://pesce.ac.in/admin/photo/6/syllabuspdf/127.pdf
PG I Year Syllabus	2024 Scheme	https://pesce.ac.in/admin/photo/6/syllabuspdf/162.pdf

4. Laboratories & Facilities

The ECE Department at PESCE, Mandya is equipped with well-established laboratories that provide hands-on experience and support various student and faculty research initiatives. The key laboratories are described below.

4.1 VLSI Design Laboratory

Established in 2010-12 with funding from the Vision Group on Science and Technology (VGST), Government of Karnataka, the VLSI Design Laboratory conducts regular training programmes for students, research scholars, and faculty. The lab is equipped with industry-standard EDA tools and FPGA development boards.

4.2 Medical and Image Processing Laboratory

This facility focuses on medical image modalities and processing, including research areas such as diabetic retinopathy, MRI image segmentation algorithms, brain disorder detection, and rheumatoid arthritis identification using thermal imaging. The lab supports faculty and doctoral research in bio-medical signal and image processing.

4.3 Signal Processing Laboratory

A dedicated Signal Processing Lab supports research and coursework in digital signal processing, communication systems, and embedded applications. Students use the lab for project work involving DSP processors, MATLAB, and hardware implementation.

4.4 Communication Systems Laboratory

The Communication Systems Lab provides facilities for experiments in analog and digital communications, microwave and antenna systems, optical fibre communication, and wireless communication. It supports both undergraduate coursework and research in 5G and next-generation communication systems.

4.5 General Infrastructure

- Well-stocked departmental library with textbooks, reference books, and journals.
- Computer centre with licensed EDA and simulation software.
- Seminar room for guest lectures, workshops, and presentations.
- Staff rooms with high-speed internet connectivity.
- VLSI and Embedded System labs for research scholars and PG students.

5. Faculty Profiles

The department has 34 teaching faculty members — including Professors, Associate Professors, and Assistant Professors — many holding Ph.D. degrees with expertise across VLSI, Signal Processing, IoT, Biomedical Engineering, Communication Systems, AI, and Machine Learning.

5.1 Professors

Dr. B S Nanda — Professor

Designation	Professor, Dept. of Electronics & Communication Engineering
Ph.D Awarded	27 July 2022 — University of Mysuru (UOM) Guide: Dr. P S Puttaswamy Thesis: Investigation on Graphene Field Effect Transistor High Frequency Flexible Transparent Displays
Research Areas	Graphene FET, Semiconductor Devices, High-Frequency Flexible Electronics, Transparent Displays
Publications	Scopus/International Journals: 04 Conferences: 02 Total: 06

Dr. Punith Kumar M B — Professor

Designation	Professor, Dept. of Electronics & Communication Engineering
Email	punithpes@gmail.com Contact: +91 98444 82308
Research Areas	VLSI Design, Signal Processing, Algorithm Architecture Design, Deep Learning for Disease Classification, Communication Systems
IEEE Role	Branch Counsellor, IEEE-PESCE Student Branch
Publications	Scopus/International Journals: 10 Conferences: 03 Total: 13
Notable Scholars Guided	Divyashree N – VLSI Algorithm for Transform Domain Coding; Vasanth Kumar Reddy G – Disease Classification in Cotton Plants using Deep Learning; Vidyashree B P – 6G Physical Layer Studies; Asha R – AI and IoMT for Biomedical Applications

5.2 Associate Professors

Dr. Mahesh — Associate Professor & Head of Department (HOD)

Designation	Associate Professor & HOD, Dept. of Electronics & Communication Engineering
Research Areas	Neuromorphic Computing, Biomedical Signal Processing, AI in Communication
Patents	"Biomimetic Neuron Model for Efficient Neuromorphic Computing" — App. No. 202441027226 (Published, co-inventor: M. Subramanyam)

Publications	Scopus/International Journals: 02 Total: 02
Ph.D Scholars	Mohan — Collision Avoidance System in Transport Industry (New Registration, Guide: Dr. Anand M J)

Dr. M Subramanyam — Associate Professor

Designation	Associate Professor, Dept. of Electronics & Communication Engineering
Research Areas	Context-Aware Systems, IoT, Neuromorphic Computing, Embedded Intelligence
Ph.D Status	Pursuing Ph.D at UOM (Guide: Dr. S S Parthasarathy) — Pre-thesis Colloquium Scheduled Thesis: Design and Implementation of Activity-Based Context-Aware System
Patents	"Biomimetic Neuron Model for Efficient Neuromorphic Computing" — App. No. 202441027226 (Published) "Integrated Framework for Context-Aware Systems: Enhancing Real-Time Response to Critical Events" — App. No. 202441027156 (Published)
Publications	Scopus/International Journals: 04 Conferences: 01 Total: 05

Kumar N Krishnamurthy — Associate Professor

Designation	Associate Professor, Dept. of Electronics & Communication Engineering
Research Areas	Image Processing, Retinopathy Detection, VLSI Signal Processing
Patents	"Development of an Algorithm for Early Detection of Plus Disease in Retinopathy of Prematurity" — App. No. 202441025734 (Published, with Dr. R Manjunatha, Dr. H S Sheshadri) "Fire Extinguishing Robot using Microcontroller" — App. No. 202441027161 (Published)
Publications	Scopus/International Journals: 10 Total: 10

Dr. M J Anand — Associate Professor

Designation	Associate Professor, Dept. of Electronics & Communication Engineering
Research Areas	IoT, Breast Cancer Detection, Electronic Nose (e-Nose), Biomedical Systems, Image Processing
Patents	"A Heuristic Approach to Develop a Prototype E-Nose to Detect Spoilage of Milk and Bread" — App. No. 202441028040 (Published) "An IoT Enabled System for Breast Cancer Detection using Image Processing" — App. No. 202241053918 (Published) "IoT Based Device to Detect the Efficiency of Remote Working using DMS Sensor" — App. No. 2020104070 (Published)
Publications	Scopus/International Journals: 15 Total: 15
Funded Project	A Heuristic Approach for Design and Development of Prototype E-Nose to Detect Spoilage of Milk and Bread — VGST (SMYSR), Karnataka — ₹5,00,000 (2016-2017, Completed 2021-22)
Ph.D Scholars	Mohan — Collision Avoidance System (New Registration at VTU)

Dr. Mahesh Gowda N M — Associate Professor

Designation	Associate Professor, Dept. of Electronics & Communication Engineering
Research Areas	Renewable Energy, Wind Power Systems, IoT, Fire Safety Robotics, AI for Environmental Classification
Patents	"Renewable Wind Power Through Modified Wind Turbine Rotor for Optimal Power Management in Smart Cities" — App. No. 2021101376 (Granted) "Fire Extinguishing Robot using Microcontroller" — App. No. 202441027161 (Published)
Publications	Scopus/International Journals: 12 Total: 12
Ph.D Scholars	Santha Kumar M T — AI for Classification and Pollutant Identification in E-Waste (New Registration, VTU)

Dr. R Manjunatha — Associate Professor

Designation	Associate Professor, Dept. of Electronics & Communication Engineering
Research Areas	Retinopathy of Prematurity (ROP), Medical Image Processing, AI and IoT Robotics
Patents	"Development of an Algorithm for Early Detection of Plus Disease in Retinopathy of Prematurity" — App. No. 202441025734 (Published)
Publications	Scopus/International Journals: 08 Conferences: 01 Total: 09
Ph.D Scholars	Yashwanth B — AI and IoT Enabled Robotic System (New Registration, VTU) Mouna K M — Risk Factors and Strategies to Prevent ROP (New Registration, VTU)

Dr. Revanesh M — Associate Professor

Designation	Associate Professor, Dept. of Electronics & Communication Engineering
Research Areas	IoT, Wireless Sensor Networks, AI-based Web Services, Blockchain, Energy-Efficient Routing
Patents	"Improving Network Lifetime: A Novel Energy-Efficient Routing Protocol for Wireless IoT Sensor Networks" — App. No. 202441028042 (Published) "An Artificially Intelligent Ontological Modelling for Consumer Centric Business Process Customization" — App. No. 202241052074 (Published) "Video Recommendations using AI" — App. No. 202241046609 (Published) "Artificial Intelligence IoT Based Wrist Band for Covid-19 Patients" — App. No. 202141003024 (Published) "IFCE-Robot Bird Flying System" — App. No. 2021100358 (Granted)
Publications	Scopus/International Journals: 14 Total: 14
Ph.D Scholars	P. Roopa Ranjini — Brain Tumour Detection (New Registration, VTU)

Dr. Sahana Raj B S — Associate Professor

Designation	Associate Professor, Dept. of Electronics & Communication Engineering
Ph.D Awarded	15 February 2023 — VTU, Belagavi Guide: Dr. V Sridhar Thesis: Study of Health Monitoring System on IoT Platform using Biometric Crypto System and Its Performance Analysis
Research Areas	IoT Health Monitoring, Biometric Cryptography, Blockchain for Healthcare, Electronic Health Records
Patents	"Blockchain Supported Searchable Encryption for Electronic Health Records" — App. No. 202022106110 (Published) "Video Recommendations using AI" — App. No. 202241046609 (Published)
Publications	Scopus/International Journals: 08 Conferences: 02 Total: 10
Award (2023-24)	Best Poster Presented at Research Conclave-2024 (PET Research Foundation & IIC, PESCE Mandya)

Dr. Vinaykumar H S — Associate Professor

Designation	Associate Professor, Dept. of Electronics & Communication Engineering
Ph.D Awarded	01 February 2025 — VTU, Belagavi Guide: Dr. H S Sheshadri Thesis: Study and Development of MR Image Enhancement Algorithms for Early Detection of Brain Disorders
Research Areas	MRI Image Enhancement, Brain Disorder Detection, Medical Image Processing, Machine Learning
Publications	Scopus/International Journals: 10 Conferences: 02 Total: 12
Award (2023-24)	Best Poster Presented at Research Conclave-2024 (PET Research Foundation & IIC, PESCE Mandya) Certificate awarded as SPOC of SWAYAM-NPTEL

5.3 Assistant Professors

Yashwanth B — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Ph.D Status	Pursuing Ph.D at VTU (Guide: Dr. R Manjunatha) Thesis: AI and IoT Enabled Robotic System for Automated Detection of HO2 (New Registration)
Publications	Scopus/International Journals: 04 Total: 04

Dr. A N Ashraya — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	Machine Learning for Medical Diagnostics, IR Thermogram Analysis
Publications	Scopus/International Journals: 04 Total: 04

Patent	"Statistical Feature Analysis of IR Thermogram Images for Rheumatoid Arthritis Detection Using Machine Learning" — App. No. 202441025734 (Published)
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Nischitha K — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Ph.D Status	Pursuing Ph.D at Adichunchanagiri University (Guide: Dr. Shobha B N) Thesis: Detection of Rheumatoid Arthritis (RA) for Knee Image Classification using Machine Learning — Course Work Completed
Research Areas	Machine Learning for Medical Image Classification, Rheumatoid Arthritis Detection
Publications	Scopus/International Journals: 06 Total: 06

Santhosh Babu K C — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Ph.D Status	Pursuing Ph.D at VTU (Guide: Dr. A M Vijay Prakash) Thesis: Performance Analysis of Low Power Digital Logic Circuits Based on FinFET Technology — Comprehensive Viva-Voce Completed
Research Areas	Low Power VLSI, FinFET Technology, Digital Logic Circuits
Publications	Scopus/International Journals: 04 Total: 04

Dr. Noor Ayesha — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Ph.D Status	Pursuing Ph.D at VTU (Guide: Dr. H S Sheshadri) Thesis: Certain Investigation on Modelling of MRI in Medical Image Computing — Thesis Completed
Research Areas	MRI Modelling, Medical Image Computing, Brain Imaging
Publications	Scopus/International Journals: 02 Conferences: 01 Total: 03

Vidyashree B P — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Ph.D Status	Pursuing Ph.D at VTU (Guide: Dr. Punith Kumar M B) Thesis: Qualitative Study of Physical Layer Aspects for Sixth Generation Wireless Communication Systems — Registered
Research Areas	6G Wireless Communication, Physical Layer Design, Signal Processing
Publications	Scopus/International Journals: 04 Total: 04

Dr. Radhika M N — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Ph.D Awarded	22 March 2022 — University of Mysuru (UOM) Guide: Dr. S S Parthasarathy Thesis: IMM Filtering for a Highly Maneuvering Target and Performance Evaluation
Research Areas	Tracking Systems, IMM Filtering, IoT Sensing, Environmental Monitoring
Publications	Scopus/International Journals: 02 Total: 02
Funded Projects	Monitoring and Detection of Ammonia and CO ₂ in Poultry Farm — KITS — ₹1,35,000 (2023-24, In Progress) Water Tank Automation — KITS — ₹85,000 (2023-24, In Progress)

Sushma P S — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	Wireless Communication, Signal Processing, Embedded Systems
Publications	Scopus/International Journals: 02 Total: 02

Spoorthy M R — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	IoT, Electronic Nose, Sensor Systems
Publications	Scopus/International Journals: 03 Total: 03
Patent	"A Heuristic Approach to Develop a Prototype E-Nose to Detect Spoilage of Milk and Bread" — App. No. 202441028040 (Published)

Chaithra K — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	Electronics, Communication Systems, Signal Processing
Publications	Scopus/International Journals: 04 Total: 04

Minugu B — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	Embedded Systems, Communication Engineering
Publications	Scopus/International Journals: 03 Total: 03

Shwetha S K — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	Electronics, VLSI, Communication

Publications	Scopus/International Journals: 02 Total: 02
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Dr. Niveditha H R — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Ph.D Status	Pursuing Ph.D at VTU (Guide: Dr. Anitha S) Thesis: Analysis and Prediction for Improving Heart Disease Detection using Machine Learning Classifier — Completed Open Seminar 2
Research Areas	Machine Learning for Heart Disease Detection, Biomedical Signal Processing
Publications	Scopus/International Journals: 01 Total: 01

Srinath M S — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Ph.D Status	Pursuing Ph.D at VIT, Chennai (Guide: Dr. R Gunabalan) Thesis: Enhancement of Charging Efficiency of Batteries for Electric Vehicles — Comprehensive Viva-Voce Completed
Research Areas	Electric Vehicle Battery Management, Power Electronics, Renewable Energy
Publications	Scopus/International Journals: 01 Total: 01
Funded Project	Portable Solar Water Bottle — KITS — ₹1,25,000 (2023-24, In Progress)

Janardhana S Y — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	Electronics, Communication Systems
Publications	Scopus/International Journals: 01 Total: 01

Yashaswini R — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	Signal Processing, Communication Engineering
Publications	Scopus/International Journals: 01 Total: 01

Sushma M P — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	Embedded Systems, IoT, Communication
Publications	Scopus/International Journals: 03 Total: 03

Divyashree H S — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	VLSI, Communication Systems, Signal Processing
Publications	Scopus/International Journals: 02 Total: 02

Vinayashree B S — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	Electronics, Communication Engineering
Publications	Scopus/International Journals: 01 Total: 01

Navya Shree Y V — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	Communication Systems, Signal Processing
Publications	Scopus/International Journals: 01 Total: 01

H S Yashvanth — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	Electronics, Embedded Systems

Safina Taj — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Research Areas	Electronics, Communication Engineering

Sanath Kumar M T — Assistant Professor

Designation	Assistant Professor, Dept. of Electronics & Communication Engineering
Ph.D Status	Pursuing Ph.D at VTU (Guide: Dr. Mahesh Gowda N M) Thesis: AI for Classification and Pollutant Identification in E-Waste (New Registration)
Research Areas	AI for Environmental Applications, E-Waste Management, IoT

5.4 Teaching Faculty – Complete Summary

#	Name	Designation
1	Dr. B S Nanda	Professor
2	Dr. Punith Kumar M B	Professor

3	Dr. Mahesh	Associate Professor & HOD
4	Dr. M Subramanyam	Associate Professor
5	Kumar N Krishnamurthy	Associate Professor
6	Dr. M J Anand	Associate Professor
7	Dr. Mahesh Gowda N M	Associate Professor
8	Dr. R Manjunatha	Associate Professor
9	Dr. Revanesh M	Associate Professor
10	Dr. Sahana Raj B S	Associate Professor
11	Dr. Vinaykumar H S	Associate Professor
12	Yashwanth B	Assistant Professor
13	Dr. A N Ashraya	Assistant Professor
14	Nischitha K	Assistant Professor
15	Santhosh Babu K C	Assistant Professor
16	Dr. Noor Ayesha	Assistant Professor
17	Vidyashree B P	Assistant Professor
18	Dr. Radhika M N	Assistant Professor
19	Sushma P S	Assistant Professor
20	Spoorthy M R	Assistant Professor
21	Chaithra K	Assistant Professor
22	Minugu B	Assistant Professor
23	Shwetha S K	Assistant Professor
24	Dr. Niveditha H R	Assistant Professor
25	Srinath M S	Assistant Professor
26	Janardhana S Y	Assistant Professor
27	Yashaswini R	Assistant Professor
28	Sushma M P	Assistant Professor
29	Divyashree H S	Assistant Professor
30	Vinayashree B S	Assistant Professor
31	Navya Shree Y V	Assistant Professor
32	H S Yashvanth	Assistant Professor
33	Safina Taj	Assistant Professor
34	Sanath Kumar M T	Assistant Professor

5.5 Technical & Non-Teaching Staff

Name	Qualification	Designation	Specialisation	Experience
Vanitha H K	B.E.	Instructor	Medical Electronics	12 Years
Madankumar H	B.E.	Tutor	Electronics & Communication Engg	9 Years
M S Kusuma	Diploma	Computer Operator	Electronics & Communication	16 Years
Meghana S P	Diploma	Instructor	Electronics & Communication	8 Years
Prathibha A	B.A.	SDA	–	10 Years
Rakshith H N	I.T.I	Mechanic	Electronics Mechanic	5 Years
Abhilash Gowda K V	Diploma	Instructor	Electronics & Communication	2 Years
Amulya T P	B.E.	Instructor	Electronics & Communication Engg	7 Months
K Y Chidananda	I.T.I	Mechanic	Electronics Mechanic	2 Years
G H Hanumanthaiah	S.S.L.C	Helper	–	16 Years
B M Shambulingaiah	PUC	Helper	–	35 Years
H Shivamahadevu	S.S.L.C	Helper	–	22 Years
Harisha V	B.A.	Helper	–	13 Years
Nandish M N	S.S.L.C	Helper	–	7 Years

6. Research — Ph.D, Publications & Patents

6.1 Ph.D Degrees Awarded

Academic Year 2024-25

Name	University	Date	Guide	Thesis Title
Vinay Kumar H S	VTU	01 Feb 2025	Dr. H S Sheshadri	Study and Development of MR Image Enhancement Algorithms for Early Detection of Brain Disorders

Academic Year 2023-24

Name	University	Date	Guide	Thesis Title
Mahesh Kumar A S	JSS	24 Feb 2024	Dr. Mallikarjunswamy M S	Study of Clinically Significant Features of Rheumatoid Arthritis (RA) using Radiographic Images & IR Thermograms
Mukul Manohar S	VTU	07 March 2024	Dr. K N Muralidhara	Video Processing and Target Tracking
Bharath M R	VTU	07 March 2024	Dr. K A Radhakrishna Rao	Multimodal Biometric Fusion for Improved Speed Security and Recognition

Academic Year 2022-23

Name	University	Date	Guide	Thesis Title
Sahana Raj B S	VTU	15 Feb 2023	Dr. V Sridhar	Study of Health Monitoring System on IoT Platform using Biometric Crypto System and Its Performance Analysis

Academic Year 2021-22

Name	University	Date	Guide	Thesis Title
Nanda B S	UOM	27 July 2022	Dr. P S Puttaswamy	Investigation on Graphene Field Effect Transistor High Frequency Flexible Transparent Displays
Radhika M R	UOM	22 March 2022	Dr. S S Parthasarathy	IMM Filtering for a Highly Maneuvering Target and Performance Evaluation

6.2 Active Research Scholars (In-Department)

Scholar	University	Guide	Research Title	Status
Noor Ayesha	VTU	Dr. H S Sheshadri	Certain Investigation on Modelling of MRI in Medical Image Computing	Thesis Completed
Divyashree N	VTU	Dr. Punith Kumar M B	Novel Efficient VLSI Algorithm for Transform Domain Coding and Decoding for Communication Application	Course Work Completed
Vasanth Kumar Reddy G	VTU	Dr. Punith Kumar M B	Early Prediction and Classification of Diseases in Cotton Plant Leaves Using Deep Learning	1st DC Meeting Completed
M Subramanya	UOM	Dr. S S Parthasarathy	Design and Implementation of Activity-Based Context-Aware System	Pre-thesis Colloquium Scheduled
Vidyashree B P	VTU	Dr. Punith Kumar M B	Qualitative Study of Physical Layer Aspects for 6G Wireless Communication Systems	Registered
Asha R	VTU	Dr. Punith Kumar M B	Development of AI and IoMT-Based System for Biomedical Applications	New Registration
Mohan	VTU	Dr. Anand M J	Collision Avoidance System in Transport Industry	New Registration
P. Roopa Ranjini	VTU	Dr. Revanesh	Brain Tumour Detection	New Registration
Yashwanth B	VTU	Dr. R Manjunatha	AI and IoT Enabled Robotic System for Automated Detection of HO2	New Registration
Mouna K M	VTU	Dr. R Manjunatha	Investigation of Risk Factors and Strategies to Prevent ROP	New Registration
Santha Kumar M T	VTU	Dr. Mahesh Gowda N M	An Application of AI for Classification and Pollutant Identification in E-Waste	New Registration

6.3 Research Scholars Pursuing Ph.D in Other Institutions

Scholar	University	Guide	Research Title	Status
K C Santhosh Babu	VTU	Dr. A M Vijay Prakash	Performance Analysis of Low Power Digital Logic Circuits Based on FinFET Technology	Comprehensive Viva-Voce Completed
Kusuma H R	REVA University	Dr. Seshikala G	Stutter Speech Detection	Course Work Completed

Nischitha K	Adichunchanagiri Univ.	Dr. Shobha B N	Detection of Rheumatoid Arthritis (RA) for Knee Image Classification using ML	Course Work Completed
Niveditha H R	VTU	Dr. Anitha S	Analysis and Prediction for Improving Heart Disease Detection using ML Classifier	Completed Open Seminar 2
Srinath M S	VIT, Chennai	Dr. R Gunabalan	Enhancement of Charging Efficiency of Batteries for Electric Vehicles	Comprehensive Viva-Voce Completed

6.4 Faculty-wise Publication Summary

Faculty Name	Scopus/Intl. Journals	Conferences	Total
Dr. K A Radhakrishna Rao	3	1	4
M. Subramanyam	4	1	5
S Jyothi	2	–	2
Dr. B S Nanda	4	2	6
Kumar N. Krishnamurthy	10	–	10
Dr. M J Anand	15	–	15
Dr. N M Mahesh Gowda	12	–	12
Dr. R Manjunatha	8	1	9
Dr. Mahesh	2	–	2
Dr. Punith Kumar M B	10	3	13
Dr. Revanesh M	14	–	14
Dr. Sahana Raj B S	8	2	10
Dr. Vinay Kumar H S	10	2	12
Ashraya A N	4	–	4
Nischitha K	6	–	6
Santhosh Babu K C	4	–	4
Mahesh Kumar A S	5	1	6
Noor Ayesha	2	1	3
Vidyashree B P	4	–	4
Dr. Radhika M N	2	–	2
Sushma P S	2	–	2
Spoorthy M R	3	–	3

Chaithra K	4	–	4
Minugu B	3	–	3
Shwetha S K	2	–	2
Niveditha H R	1	–	1
Yashwanth B	4	–	4
Janardhana S Y	1	–	1
Yashaswini R	1	–	1
Sushma M P	3	–	3
Divyashree H S	2	–	2
Vinayashree B S	1	–	1
Navyashree Y V	1	–	1

6.5 Patents

Author(s)	Patent Title	App. No.	Status
Dr. R Manjunatha, Dr. H S Sheshadri, Kumar N Krishna Murthy	Development of an Algorithm for Early Detection of Plus Disease in Retinopathy of Prematurity	202441025734	Published
Maresh Kumar A S, Dr. Punith Kumar M B, Ashraya A N, Nischitha K, Vinay Kumar H S	Statistical Feature Analysis of IR Thermogram Images for RA Detection Using Machine Learning	202441025734	Published
Dr. Mahesh, M. Subramanyam	Biomimetic Neuron Model for Efficient Neuromorphic Computing	202441027226	Published
Dr. Revanesh M, Dr. Sahana Raj B S, Santhosh Babu, Ashraya A N, Vidyashree B P	Improving Network Lifetime: A Novel Energy-Efficient Routing Protocol for Wireless IoT Sensor Networks	202441028042	Published
M Subramanyam, Dr. S S Parthasarathy	Integrated Framework for Context-Aware Systems: Enhancing Real-Time Response to Critical Events	202441027156	Published
Dr. M J Anand, Dr. H R Divakar, Spoorthy M R	A Heuristic Approach to Develop a Prototype E-Nose to Detect Spoilage of Milk and Bread	202441028040	Published
Dr. N M Mahesh Gowda, Kumar N. Krishnamurthy, S Jyothi, Dr. Nanda B S, Noor Ayesha	Fire Extinguishing Robot using Microcontroller	202441027161	Published
Dr. B S Sahana Raj	Blockchain Supported Searchable Encryption for Electronic Health Records	202022106110	Published
Dr. M Revanesh	An Artificially Intelligent Ontological Modelling for Consumer Centric Business Process Customization	202241052074	Published

Dr. B S Sahana Raj, Dr. M Revanesh	Video Recommendations using AI	202241046609	Published
Dr. M J Anand	An IoT Enabled System for Breast Cancer Detection using Image Processing	202241053918	Published
Mr. Mahesh Kumar A S	A System for Development of Non-Invasive Prenatal Testing (NIPT) Model Using AI & ML	202141033346	Published
Mr. Mahesh Kumar A S	Development of Multi-Sensor Smart Robot for Intelligent Border Security Surveillance System	202141033795	Published
Mr. Mahesh Kumar A S	Machine Learning and IoT Based Smart Wearable System for Autonomous Management of Diabetes Mellitus	202141030571	Published
Mr. Mahesh Kumar A S	IoT and RFID Based Optimal Positioning of Autonomous Shock Absorber for Smart Vehicles	202141028518	Published
Mr. Mahesh Kumar A S	Machine Learning Based Smart Grid Power System Integrated with Wireless Communication Networks	202111027550	Published
Mr. Mahesh Kumar A S	Artificial Intelligence Based Autonomous Pest Detection in Smart Agricultural Farms	202121026289	Published
Mr. Mahesh Kumar A S	Thermal Camera for Detecting Rheumatoid Arthritis	382293-001	Published
Mr. Mahesh Kumar A S	NLP & AI Based Automatic Detection of Infection Rate of Covid-19	202111023806	Published
Mr. Mahesh Kumar A S	Intelligent Battery Intel Swapping Technology for E-Vehicle	202141023431	Published
Mr. Mahesh Kumar A S	Machine Learning Based EEG Signal Processing for Smart Patient Monitoring System	202141022214	Published
Mr. Mahesh Kumar A S	Machine Learning Based Breast Cancer Detection by Neuro Fuzzy Logic	202141018245	Published
Mr. Mahesh Kumar A S	Machine Learning Based Battery Management System for Optimized EV Performance	202141017715	Published
Mr. Mahesh Kumar A S	Artificial Intelligence Based Cardiac Arrhythmia Detection in Heart Patients	202121010624	Published
Dr. Revanesh M	Artificial Intelligence IoT Based Wrist Band for Covid-19 Patients	202141003024	Published
Dr. N M Mahesh Gowda	Renewable Wind Power Through Modified Wind Turbine Rotor for Optimal Power Management in Smart Cities	2021101376	Granted
Dr. M Revanesh	IFCE-Robot Bird Flying System: Intelligent Robot Bird Flying Controllability Evaluation System	2021100358	Granted

Dr. M J Anand	IoT Based Device to Detect the Efficiency of Remote Working using DMS Sensor	2020104070	Published
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6.6 Funded Research Projects

Principal Investigator	Project Title	Year	Funding Agency	Amount (Rs.)	Status
Dr. Radhika M N	Monitoring and Detection of Ammonia and CO ₂ in Poultry Farm	2023-24	Karnataka Innovation & Technology Society (KITS)	1,35,000	In Progress
Dr. Radhika M N	Water Tank Automation	2023-24	Karnataka Innovation & Technology Society (KITS)	85,000	In Progress
Srinath M S	Portable Solar Water Bottle	2023-24	Karnataka Innovation and Technology Society (KITS)	1,25,000	In Progress
Dr. M J Anand	Heuristic Approach for Design and Development of Prototype E-Nose to Detect Spoilage of Milk and Bread	2016-2017	VGST (SMYSR), Karnataka	5,00,000	Completed (2021-22)
Dr. H S Sheshadri	Establishment of Medical Image Analysis Laboratory	2014-2015	VGST	20,00,000	Completed
Dr. V Sridhar	Infrastructure Strengthening in Science and Technology in Higher Educational Institutions (K-FIST) – Level 1	2010-2012	VGST	20,00,000	Completed

7. Placements

The ECE department has a commendable placement record, with students consistently being recruited by leading IT and core engineering companies. The department's Training and Placement Cell actively facilitates campus recruitment drives.

Academic Year	Total Students	Total Offers Received	Students Placed	Students for Higher Studies
2024-25	201	98	98	–
2023-24	167	115	115	5
2022-23	196	149	149	4

8. Student & Faculty Achievements

8.1 Academic Year 2024-25 — Student Achievements

Student Name(s)	Achievement
Vikas S, Yashvanth H T, Gowtham M, Kushal G M	Second prize in IDEATHON-2k24 organised by BGS Institute of Technology, Adichunchanagiri University
Jashwanth Gowda S P, Chandan N, Yashwanth V Rai, Nandish D S	₹10,000 cash prize in National Level Hackathon at GSSSIETW, Mysuru
Ananya (4PS23EC403), Preethi K M (4PS22EC122)	Second prize in IDEATHON organised by BGSIT AICTE IDEA Lab
Vishwas S, Akash B, Nikhilesh G, Srinivasa H S, Suman M L	Winners of Inter-Collegiate Mysuru Division Level KHO-KHO Tournament at MIT, Thandavapura
Tejas M S	Bronze medal in Low Kick — IAKO National Kickboxing Championship 2024 at V.K. Krishnamenon Indoor Stadium, Kozhikode, Kerala

8.2 Academic Year 2023-24 — Faculty Achievements

Faculty Name	Achievement
Dr. Vinay Kumar H S	Best Poster Presented at Research Conclave-2024 organised by PET Research Foundation in collaboration with IIC, PESCE Mandya
Dr. Vinay Kumar H S	Certificate awarded as SPOC of SWAYAM-NPTEL

8.3 Academic Year 2023-24 — Student Achievements

Student Name(s)	Achievement
Pradeep Nagavath	3rd place in Plasma Pull — Annual Technical Fest of IIT Roorkee
Harsh Rajkumar	1st place in Pitch Premier — Techkriti'24, IIT Kanpur
Usha B M	10th Rank in Java Programming Quiz — PVPSIT, Vijayawada
Arun Kumar K S	Best Idea Award — Innovative Idea Competition, Vidyavardhaka College of Engineering, Mysuru
Preethamraj R	First prize in Entrophosis — Pragyatha'23 Hackathon, MCE Hassan
Mahentesh, Harshith S	Runner-up in Badminton — Inter-Collegiate Mysuru Division Level Tournament, NIE South Campus, Mysuru
Chethan Kumar K	Runner-up in Kabaddi — VTU Inter-Collegiate Mysuru Division Level Tournament, Mysore College of Engineering

Vishwas Gowda C H, Chethan Kumar K	Runner-up in Kabaddi — VTU Mysuru Zone Kabaddi Championship
Aniruddh P Koundinya	Awarded IEEE Outstanding Young Professionals Volunteer

8.4 Academic Year 2022-23 — Student Achievements

Student Name(s)	Achievement
Pratiksha Y	Participated in Project Review at Sir M Visvesvaraya Institute of Technology, Bangalore
Skanda M K	Awarded IEEE Bangalore Section Chair's Recognition Award
Manikant Amaresh Savadatti	Won 2nd prize in Dare to Dream Innovation Contest 3.0 by DRDO, Ahmedabad

9. Professional Bodies & Student Clubs

9.1 IEEE PESCE Student Branch

The IEEE-PESCE Student Branch, a division under IEEE Region 10, was established in 2007. It has grown to 100+ active members and organises a range of activities including technical events, seminars, workshops, and social activities. The branch has received multiple awards from the IEEE Mysore Subsection.

Faculty Contacts

Branch Mentor	Dr. Vinay S vinay@pesce.ac.in +91 9986515835
Branch Counsellor	Dr. Punith Kumar M B punithpes@gmail.com +91 98444 82308
WIE Society Mentor	Dr. M C Padma padmamc@pesce.ac.in +91 9141113030
COMSOC Mentor	Dr. Anitha M L anithaml@pesce.ac.in +91 9945576186
Student Chairperson	Pavan Kumar K pavankumar8656@gmail.com +91 7603858545
Official Website	http://www.ieeepesce.com

Awards Won by IEEE-PESCE Student Branch

Award	Year	Recipient
IEEE Mysore Subsection Outstanding Student Volunteer Award	2023	Aniruddh P Koundinya
IEEE Mysore Subsection Most Promising Student Branch Award	2021	PESCE Student Branch
IEEE Mysore Subsection Outstanding Student Volunteer	2021	Yeshwant L M
IEEE Mysore Subsection Outstanding Student Volunteer	2021	Skanda M K

IEEE-Funded Projects

Project Name	Developed By	Funded By
Virobot: The Artificial Nurse Project	Manikanth Amaresh and Savadatti	IEEE India Council
Virobot Mini Prototype	Manikanth Amaresh, Savadatti, Sushma H S, Dhanush J, Preetham Kishore	IEEE Student Humanitarian Project
Internet Multi Bot	Manikanth Amaresh, Savadatti, Anup P H, Sushmitha H S, Bhoomika H J	IEEE Student Innovative Project

9.2 ISTE PESCE Chapter

PES College of Engineering became an Institutional Member of ISTE in 1986. The ISTE-PESCE Chapter was established in 1989. All staff members are life members of ISTE. The chapter conducts short-term courses, workshops, and the annual Eureka programme.

ISTE President	Dr. Veena M N, Ph.D. – Professor, Dept. of MCA Mob: +91 98451 42764
ISTE Secretary	Dr. Nayaka S R, Ph.D. – Asst. Professor, Dept. of Mathematics Mob: +91 96116 00098
Facebook	https://www.facebook.com/pesceiste/
Instagram	https://www.instagram.com/pesce.iste/

10. Contact Information

Department	Electronics & Communication Engineering (ECE)
Head of Department	Dr. Mahesh – Associate Professor & HOD
Department Page	https://pesce.ac.in/dep-electronics-communication-engg.php
College Address	K V Shankaragowda Rd, PES College Campus, Mandya, Karnataka – 571 401
Email	admissions@pesce.ac.in
Mobile	+91 9448282588
Office Phone	08232 – 220043
Website	https://pesce.ac.in
IEEE Branch	http://www.ieeeesce.com

10.1 Quick Links

Resource	Link
Department Page	https://pesce.ac.in/dep-electronics-communication-engg.php
Admissions	https://pesce.ac.in/admissions.php
NIRF Data	https://pesce.ac.in/nirf.php
NAAC Certificate	https://pesce.ac.in/naac.php
Research / Ph.D	https://pesce.ac.in/phd.php
Placements	https://pesce.ac.in/placement.php
Alumni	https://pesce.ac.in/alumni.php
IQAC	https://pesce.ac.in/footer-internal-quality-assurance-cell.php
AICTE Idea Lab	https://pesce.ac.in/aicte_Idea_lab.php
IIC	https://pesce.ac.in/iic.php
Grievance	https://pesce.ac.in/committees.php
Direct Verify	https://pesce.ac.in/directverify.in
PES ERP	https://pesgroup.dhi-edu.com
Centre of Excellence	https://pesce.ac.in/img/departement/Electronics/Centre_of_Excellence.pdf